UrbanTwin-AI

Geospatial Intelligence and Traffic Optimization for Sustainable Cities

Independent research prototype | Cairo area of interest

Research question

Can spatial indicators derived from Earth observation be analyzed alongside controlled traffic-signal simulations to support reproducible urban research?

Scope and integrity

The satellite and ESA WorldCover measurements refer to a small Cairo study area. The repository's default SUMO intersection is synthetic, not a reconstruction of Cairo's street geometry, measured traffic flows, or emissions. Spatial and simulation findings are presented separately.

Methods

Sentinel-2 L2A imagery is SCL-screened and aligned in a projected CRS, then compared with ESA WorldCover 2021 reference labels. Random Forest and histogram gradient boosting use blocked spatial holdouts. GA and PSO tune two SUMO signal green durations using training seeds; held-out seeds evaluate performance against a fixed-time baseline.

Experimental status

METHODOLOGY-ONLY DRAFT: no real satellite-model results have been generated in this repository. Do not cite this draft as an experimental finding.

SUMO status

No SUMO benchmark results recorded yet.

Limitations and future work

Single AOI, scene-cloud filtering, WorldCover pseudo-labels, spatial correlation between adjacent blocks, limited simulation seeds, and abstract traffic demand constrain generalization. Future work can integrate a validated real-road network and observed travel-demand counts.

Data attribution

Contains Copernicus Sentinel data; ESA WorldCover © ESA; road data © OpenStreetMap contributors (ODbL) when used. See repository documentation for data sources and licensing.